

Crypto Toolkit Project

Create a website or application with a graphical user interface (GUI) that allows users to enter text messages and apply different cryptographic operations.

Required Features

1. Symmetric Encryption and Decryption

The application should support the following algorithms:

- AES
- DES
- 3DES

For each algorithm, the user should be able to encrypt and decrypt text using a secret key.

2. Asymmetric Encryption and Decryption

The application should support the following algorithm:

- RSA

The user should be able to generate keys, encrypt text, and decrypt it.

3. Encoding and Decoding

The application should support the following techniques:

- Base64
- Hex
- URL Encoding

The user should be able to encode and decode text.

4. Hashing

The application should support the following algorithms:

- SHA-256
- SHA-512
- Salted Hashing

Project Requirements

Each team should submit:

- A working website or application with a simple and clear GUI.
- Source code.
- A PDF report.
- Screenshots showing the main features of the application.
- A ZIP file containing both the report and the source code.

Report Content

The report should briefly include the following sections:

1. Algorithms Used

Briefly explain the algorithms and techniques implemented in the application, including:

AES, DES, 3DES, RSA, Base64, Hex, URL Encoding, SHA-256, SHA-512, and Salted Hashing.

2. Implementation Screenshots

Add screenshots from the application showing:

- Encryption
- Decryption
- Encoding
- Decoding
- Hashing

3. Testing Examples

Add at least one example for each feature

Each example should include the input, the selected operation, and the output.

Evaluation Criteria

- Correct implementation of encryption and decryption: 30%
- Encoding, decoding, and hashing features: 20%
- Report and screenshots: 20%
- Discussion and understanding: 30%

Submission Details

- Team members: 4 students
- Team submission deadline: 13 May
- Team submission form: <https://forms.gle/DxUvffJJFvsMZLpRA>
- Project submission deadline: 16 May at 11:59 PM
- Project submission : <https://forms.gle/AtA3zXFfe2mGXLoB6>
- Upload a ZIP file containing the report and the source code.